

Linh Tran

<https://lnhtrn.github.io/>

Rochester, NY, USA

lt297@njit.edu

Education	New Jersey Institute of Technology Ph.D. in Information Systems <i>Specialization: Human-Computer Interaction</i> Jan 2025-Dec 2025
	University of Rochester (GPA: 4.0) Masters of Science in Computer Science <i>Specialization: Artificial Intelligence and Human-Computer Interaction</i> Courses: Human Computer Interaction, Introduction to Augmented and Virtual Reality, End-to-End Deep Learning, Machine Vision, Algorithmic Game Theory Jan 2025-Dec 2025
	University of Rochester Bachelor of Science in Computer Science, Minor in Mathematics Relevant Courses: Machine Learning, Natural Language Processing, Data Structures & Algorithms, Computer Organization. Aug 2020-Dec 2024
Publication	S. Chatterjee, L. Tran , Q. D. Nguyen, R. Kirson, D. Hamlin, H. Aquino, H. Lyu, J. Luo, and T. Dye. <i>Assessing Historical Structural Oppression Worldwide via Rule-Guided Prompting of Large Language Models</i> . To appear in IEEE International Conference on Big Data (IEEE BigData 2025).
Poster Presentations	<i>LLMs as Sociologists: Leveraging AI/ML contextual knowledge for Social Oppression</i> . 2025 Finger Lakes Science and Technology Showcase, Rochester, NY <i>Investigation of Racial Bias in Vision-Language Assistants in Workplace Settings</i> . ACM New York Celebration of Women in Computing 2025 (WiCNYC '25). <i>Can Finetuning Overcome Racial Bias In Face Recognition Models?</i> University of Rochester Schwartz Discover Grant Symposium, 2024.
Research Experience	ROC-HCI BEAR Lab with Dr. Yukang Yan <i>AR Housework Assistant for Individuals with Executive Dysfunction</i> Feb 2025–Present - Enabled context-aware task prompting for individuals with ADHD by integrating real-time object recognition and LLM into Snap Spectacles to reduce executive dysfunction when planning household chores using TypeScript. - Researched and planned a formative user study to evaluate usability and impact of executive function on completing household chores and how to mitigate. - Trained and deployed a custom YOLOv9 object detection model to recognize household items (e.g., dishes, laundry) as part of an AR system built with Unity and C# for Meta Quest 3. <i>Adaptive DR system that automatically detects and removes distracting objects in real-time</i> - Developing automatic drift calibration for Meta Quest's Passthrough real time camera through the use of QR codes using C#. <i>Computer Vision for dynamic spatial placement of AR/VR applications in changing environments</i> - Developed user study questionnaire and conducted pilot studies with participants to evaluate interaction preferences for VR application locations. Dye Lab, University of Rochester Medical Center with Dr. Tim Dye Data Science Research Assistant Sep 2024–Present - Collaborated with epidemiologists to develop a LLM-based tool that analyzes global COVID-19 survey responses for indicators of systemic oppression and demographic bias. - Engineered 20+ text prompts to elicit nuanced responses from large language models (Phi-4, Gemma, DeepSeek), focusing on social inequity, discrimination, and marginalization per country. - Implemented a scoring system based on token probabilities using PyTorch and Transformers to classify and rank LLM responses based on severity and specificity of oppressive language.

- Enhanced model contextuality by integrating few-shot learning using curated country-specific knowledge and sociopolitical data to improve sensitivity to cultural and national variations.

kLab, University of Rochester with Dr. Christopher Kanan Jan 2024–Mar 2025

Bias Detection and Mitigation in Vision-Language and Computer Vision Models

- Designed and implemented a bias evaluation framework using Python and PyTorch to analyze over 20 vision models (e.g., ResNet, VGG-13, DINOv2), achieving a 9% reduction in demographic bias while maintaining 80% task accuracy.
- Investigated racial bias in corporate hiring pipelines by prompting Vision-Language Models (VLMs) to evaluate candidate traits (e.g., professionalism, competence) using only facial images, revealing biased patterns that may disadvantage marginalized applicants.
- Developed a GPU-accelerated CI/CD pipeline with Weights & Biases to automate the testing, logging, and deployment of deep neural networks in fairness-sensitive evaluation tasks.

Work
Experience

Bryan R. Harrison, PhD Psychologist, PC

June 2025–Aug 2025

Machine Learning/AI Developer Intern

- Designed and deployed a web-based assistive tool that streamlined clinician workflows, increasing efficiency by 200% by transforming speech into structured reports via Python, OpenAI API, and Streamlit.
- Developed human-centered prompt-engineering strategies and website interface, enabling clinicians to interact naturally with AI systems without technical expertise.
- Conducted contextual inquiry interviews, iterative user testing and feedback sessions with clinicians to refine usability, accessibility, and alignment with healthcare tasks.
- Implemented secure Google OAuth authentication and containerized the application with Docker on Google Cloud Platform for scalable, HIPAA-compliant deployment.

VinBigData

May 2023–Aug 2023

Data Science Intern

- Optimized a large language model (LLM) question classifier and enhanced its user-facing accuracy, improving the experience for end-users interacting with automated Q&A systems.
- Augmented the SQuAD 2.0 dataset with 50,000 high-quality question-answer pairs to reduce false answers, support better human-AI comprehension and interaction.
- Deployed a multilingual semantic analysis model via TorchServe, automating post-processing and accelerating accuracy evaluation speed by 200%.
- Collaborated with research scientists and ML engineers to enhance production-readiness and usability of NLP pipelines.

HADTech Joint Stock Company

Feb 2021–May 2021

Web Development Intern & Frontend Team Leader

- Collected and analyzed user feedback to inform website and app design, ensuring alignment with customer needs and usability best practices.
- Designed key website components, including dashboards, search, and profile pages, using Figma to create intuitive and visually coherent interfaces.
- Developed and prototyped mobile app and web interfaces, iteratively refining designs based on team and user input.
- Collaborated with cross-functional teams to integrate design with frontend and backend systems, balancing usability with technical feasibility.

Projects

ADHD-Assist: AR Housework Assistant for Executive Dysfunction Sep 2025–Dec 2025

Project Demo (GitHub) - Continuation of Research Project

- Designed and implemented an augmented reality housework assistant to support individuals with ADHD and executive dysfunction, using spatial UI to deliver context-aware task guidance directly within real-world environments.
- Designed and prototyped user interface to gamify household chores as game quests with simple directives, allowing users to easily start and finish tasks by alleviating mental blocks created by executive dysfunction.

- Developed a vision-based pipeline that converts room screenshots into automatically generated chore lists using OpenAI API and TypeScript, integrating mixed reality interaction design with simple AI-driven task structuring.
- Built and evaluated a wearable AR prototype on Snap Spectacles, gaining experience in mixed reality systems, human-centered interface design, and creating calm, non-overstimulating user experiences.

Teaching Assistantship	CSC 246 Machine Learning, <i>Teaching Assistant</i>	Spring 2024, Fall 2024
	CSC 242 Introduction to Artificial Intelligence, <i>Teaching Assistant</i>	Fall 2023
	The Learning Center, <i>Tutor, Study Group Leader, Study Zone Leader</i>	Aug 2023–Present
Community Involvement	University of Rochester Orientation, <i>Orientation Week Leader</i>	Aug 2024–Sep 2024
	Vietnamese Student Association, <i>Event Manager, Executive Board</i>	Sep 2022–Sep 2023
	Lilac Housing, <i>Vice President, Executive Board</i>	Sep 2021–May 2022